M_GIM User Manual

Multi-system global and regional ionospheric modeling software

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1. Introduction

As one of the main error sources that Global Navigation Satellite Systems (GNSS) suffer from, the ionospheric delay is necessary to be modeled and analyzed. This User Manual introduces M_GIM, a multi-system global and regional ionospheric modeling software. The M_GIM software is based on the M_DCB (Jin et al., 2012)—a GNSS satellite and receiver differential code biases estimating software—and redeveloped in Matlab 2019b. The conventional dual-frequency carrier-to-code leveling (DFCCL) method is used in M_GIM, and up to quad-system (GPS/GLONASS/Galileo/BDS) observations can be processed. Using this software, one can easily establish a reliable temporal and spatial variation model of the ionosphere.

2. Mathematical models

This section starts with a review of the DFCCL method to extract STEC, proceeds to several ionospheric models, then follows the sequential adjustment data processing strategy, and ends with an introduction of the methods of ionospheric modeling and root-mean-square (RMS) mapping.

2.1 STEC extracting

Since the ionospheric delay depends on the frequency of the observables, the first-order delay can be isolated with dual-frequency pseudorange or carrier phase measurements (Hauschild, 2017), and high order ionospheric refraction effects are usually ignored, which account for less than 0.1% of the ionospheric delay (Hernandez-Pajares et al., 2014). The Geometry-Free (GF) combination of dual-frequency pseudorange or carrier phase measurements can be expressed as:

$$P_4^S = P_{1,j}^{i,S} - P_{2,j}^{i,S} = \left(I_{1,j}^{i,S} - I_{2,j}^{i,S}\right) + c \cdot DCB^{i,S} + c \cdot DCB_j^S \tag{1}$$

$$L_4^S = L_{1,j}^{i,S} - L_{2,j}^{i,S} = -\left(I_{1,j}^{i,S} - I_{2,j}^{i,S}\right) - \lambda\left(b_{1,j}^{i,S} - b_{2,j}^{i,S}\right) - \lambda\left(N_{1,j}^{i,S} - N_{2,j}^{i,S}\right) \tag{2}$$

where P_4^S and L_4^S denote the pseudorange and carrier phase observables of the GF combination, $P_{1,j}^{i,S}$ and $L_{1,j}^{i,S}$ denote the code and phase observables on frequency 1, with *j*, *i*, *S* denoting the receiver, satellites number and satellites system respectively; $I_j^{i,S}$ is the ionospheric delay along the line of sight, $DCB^{i,S}$ and DCB_j^S refer to the DCB of the satellites and receivers for different satellite systems; $N_j^{i,S}$ is the ambiguity parameter, and $b_j^{i,S}$ is the phase instrumental delays, which is commonly absorbed by ambiguity parameters; *c* is the speed of light, and λ is the wavelength.

Since pseudorange multipath effects and thermal noises are significantly amplified in pseudorange-based STEC calculation (Li et al., 2020), the DFCCL method is commonly used to get a more accurate STEC estimate. In one constant arc, more smoothed P_4^S observations, $\tilde{P}_{4,t}^S$, can be obtained through L_4^S smoothing.

The magnitude of ionospheric delay correction caused by electromagnetic waves in the propagation medium can be expressed as:

$$I_{k,j}^{i,S} = \frac{40.28}{f_k^2}\int Ne \cdot ds = \frac{40.28}{f_k^2} STEC \tag{3}$$

where f_k denotes the carrier phase frequency on frequency k; Ne denotes the electron density in the path s; $\int Ne \cdot ds$ means the integral extended along the path of the signal, which can be substituted by *STEC*. The STEC is given in TEC units (TECU), where $1\ \text{TECU} = 10^{16}\ electrons\ per\ m^2$.

Substituting Eq. (3) into Eq. (1), an equation related to STEC observations can be written as:

$$\tilde{P}_{4,t}^{S} = \frac{40.28 \cdot (f_2^{S^2} - f_1^{S^2})}{f_1^{S^2} \cdot f_2^{S^2}} STEC^S + c \cdot DCB^{i,S} + c \cdot DCB_j^S \tag{4}$$

which can be seen as a preliminary form of the observation equations.

2.2 Ionospheric model

Assuming that all free electrons are gathered in a shell of infinitesimal thickness at altitude H, the single-layer model is adopted to simplify the ionospheric model, where H corresponds to the altitude of maximum electron density. Every signal between satellite and receiver will produce a penetration point on the single-layer, called the ionosphere piercing point (IPP), and the single-layer model is commonly used in converting slant ionospheric delay to the zenith path along each line of sight, which can be written as (Schaer, 1999):

$$VTEC^S = F\left(z_j^{i,S}\right) \cdot STEC^S \tag{5}$$

$$F\left(z_j^{i,S}\right) = \cos\left(\arcsin\left(\frac{R}{R+H}\sin\left(z_j^{i,S}\right)\right)\right) \tag{6}$$

Where $F(z_j^{i,S})$ is the mapping function, and $z_j^{i,S}$ denotes the zenith distance at the receiver; $R = 6371$ *km* for the mean radius of the earth, and $H = 450$ *km* for the height of the assumed single layer, and both of which are consistent with the value used by the CODE group (Unbenanntes Dokument (unibe.ch)).

For different satellite systems, $\frac{f_1^{S^2} \cdot f_2^{S^2}}{40.28 \cdot (f_2^{S^2} - f_1^{S^2})}$ can be seen as a common constant associated with the dual-frequency, replaced by k^S. Substituting Eq. (5) into Eq. (4), the final observation equations for different systems can be rewritten as:

$$k^S \cdot F\left(z_j^{i,S}\right) \cdot \tilde{P}_{4,t}^{S} = VTEC^S + k^S \cdot F\left(z_j^{i,S}\right) \cdot \left(c \cdot DCB^{i,S} + c \cdot DCB_j^S\right) \tag{7}$$

2.2.1 Spherical harmonic model

Previous research has proved that the spherical harmonic function is suitable for ionospheric vertical total electron content (VTEC) modeling on a global scale (Schaer, 1999; Li et al., 2015; Li et al., 2020), which can be expressed as:

$$VTEC(\beta, \lambda) = \sum_{n=0}^{n_{max}} \sum_{m=0}^{n} \widetilde{P_{nm}}\,(\sin\beta)(A_{nm}\cos m\lambda + B_{nm}\sin m\lambda) \tag{8}$$

where $VTEC(\beta, \lambda)$ denotes the global VTEC value of the IPP at latitude β and longitude λ in the

sun-fixed geomagnetic frame; *n* and *m* denote the order and degree of the spherical harmonic function; $\tilde{P}_{nm}$ is the normalized associated Legendre function; A_{nm} and B_{nm} are the normalized ionospheric model parameters, which need to be estimated in the global ionospheric modeling.

2.2.2 Polynomial model

Different from global ionospheric modeling, the orthogonality of the spherical harmonic model is lost in regional ionospheric modeling. To fit the characteristics of regions better, the polynomial model is proposed, which can be regarded as a function of longitude and latitude differences in the sun-fixed geomagnetic frame. The polynomial model can be expressed as (Li and Huang, 2005):

$$VTEC(\beta,\lambda) = \sum_{i=0}^{n}\sum_{j=0}^{m} E_{ij}(\beta-\beta_0)^i(\lambda-\lambda_0)^j \tag{9}$$

where β_0 and λ_0 denote the latitude and longitude of the geometric center in the region; β and λ denote the latitude and longitude of the IPP, and all of them are in the sun-fixed geomagnetic frame; *n* and *m* denote the max order and degree of the polynomial model; E_{ij} denotes the regional ionospheric model parameters, which should be estimated in the regional ionospheric modeling.

2.2.3 Nonintegral spherical harmonic model

Considering the irregularity of the regional ionosphere, a nonintegral spherical harmonic model is proposed based on the spherical model, which can be regarded as a general form of the latter, and written as:

$$VTEC(\beta,\lambda) = \sum_{i=0}^{n}\sum_{j=0}^{m} \widetilde{P_{nm}}(\cos\beta)(A_{nm}\cos m\lambda + B_{nm}\sin m\lambda) \tag{10}$$

where *n* and *m* denote the max order and degree of the nonintegral spherical harmonic model, with *m* commonly smaller than *n*, and *m* not larger than *i*. A_{nm} and B_{nm} are the model parameters that need to be estimated similarly.

2.3 Sequential adjustment

Since the ionosphere is a highly varying medium, and the stations on the earth can get the observations continuously, a sequential estimation scheme is commonly adopted (Feltens, 2007), to reduce the storage loss of the computer greatly. Simplified the observation equation of Eq. (7), the equations for an individual system *S* can be expressed as:

$$v^S = B^S\hat{x} - l^S \tag{11}$$

For ionosphere modeling in M_GIM, $\hat{x}$ is formed as $\left[DCB^{i,S} \quad DCB_j^S \quad IONC\right]$, denoting the satellite DCBs, receiver DCBs and ionospheric modelling parameters respectively; B^S denotes the design matrix of parameters; l^S and v^S denote the real observations and observation residuals. Due to the linear correlation between the satellite and receiver DCBs, it is not feasible to separate them directly. To eliminate the rank-deficiency, a zero-mean condition on all the satellite DCBs of each system *S* is imposed, which can be expressed as:

$$\sum_{i=0}^{n^S} DCB^{i,S} = w^S \tag{12}$$

where n^S denotes the satellite number, w^S denotes the negative value of the sum of missing satellite DCB, which satisfies $w^S = 0$ when no satellites are missing. Correspondingly, $G^{S^T}\hat{x} = w^S$ will be added into the observation equations as a constraint condition.

For reducing the storage loss, sequential adjustment is used in the multi-system data processing, and the final result can be calculated as:

$$\hat{x} = \left(\sum_S B^{S^T} P^S B^S + \sum_S G^S G^{S^T}\right)^{-1} \cdot \sum_S (B^{S^T} P^S l^S + G^S w^S) \tag{13}$$

where P^S denotes the weight matrix of the observations.

2.4 Methods of ionospheric modeling and RMS mapping

The estimated standard deviation m_0 can be computed as (Schaer, 1999):

$$\begin{cases} m_0 = \sqrt{\dfrac{v^{S^T} P^S v^S}{f}} \\ v^{S^T} P^S v^S = l^{S^T} P^S l^S - \hat{x}^T \cdot B^{S^T} P^S l^S \\ f = t - u \end{cases} \tag{14}$$

where *f* denotes the degree of freedom (DOF) of the least-squares (LS) adjustment, *t* and *u* denote the number of observations and adjusted model parameters. Finally, the ionospheric VTEC *y* can be calculated as:

$$y = A\hat{x} \tag{15}$$

where $A = f(ep, \beta, \lambda)$ means the ionospheric coefficients calculated by the ionospheric model in epoch time *ep*. Then, the final RMS can be calculated as:

$$\begin{cases} m_y = m_0 \cdot \sqrt{Q_{yy}} \\ Q_{yy} = A Q_{\hat{x}\hat{x}} A^T \\ Q_{\hat{x}\hat{x}} = \left(B^{S^T} P^S B^S\right)^{-1} \end{cases} \tag{16}$$

where m_y denotes the formal variances, commonly multiplied by a factor such as 10 due to the conservative estimating; Q_{yy} denotes the covariance matrices; $Q_{\hat{x}\hat{x}}$ denotes the cofactor matrix of the model parameters.

3. M_GIM Software

3.1 Example

Example input data are provided in a separate download available on the GPS Toolbox website and also at: https://mega.nz/file/yIokCBiL#gjKCcKIMcX2zeSw6uTMB0z50TofmMrbQ47c15NbI2Ak, 1.85 Gb). To run the examples, the data contained in this download should be substituted into the corresponding M_GIM folders. The M_GIM software distribution includes a M_Result_From_Test_Data folder that contains the final results of the three main routines executed using the Test_Data input files. Users can run the examples and compare their results to M_Result_From_Test_Data to verify software performance.

3.2 Dependencies

The M_GIM software is packaged as a Matlab toolbox. Its functionality requires the following software, which is included with the M_GIM distribution on the GPS Toolbox website:

1. M_Map (Pawlowicz, 2020), a mapping toolbox
2. M_DCB (Jin et al., 2012), a MATLAB-based software for estimating DCBs

3.3 Data preparation

Rinex 3.01 or higher format observation files are recommended in M_GIM. Multi-system precise ephemeris files in SP3 format and DCB files in BSX format are also needed in data processing. The relevant data can be downloaded from the official organizations, refer to the websites listed in the appendix for details.

In the data preprocessing stage, the downloaded observation data needs to be processed in two steps: file decompression and format conversion. Observation files with suffixes *.crx and *.d are converted to *.rnx and *.o files by using the **crx2rnx** software, which can be found in the **Tools** folder, and a brief using method can refer to the attached guide. Since some observation data provided by IGS stations are in RINEX 2 format, it is necessary to use the **gfzrnx** software to convert to RINEX 3 format. The **gfzrnx** software is developed by the German Research Centre for Geosciences (GFZ), and users can process the observation files according to the guidance provided.

Before running M_GIM, the necessary data files must be placed into specific directories in the /M_GIM folder. Table 1 details the folder names and data types that should be placed in each folder.

Table 1 Subdirectories of /M_GIM, with the required input files for a modeling run

FOLDER	DESCRIPTION OF FILES
	INPUT
Files_rinex/global	Global observation files, in RINEX 3 format. (uncompressed)
Files_rinex/regional	Regional observation files, in RINEX 3 format. (uncompressed)
Files_SP3	Precision ephemeris files, in SP3 format. For multiple GNSS processing, the rapid precision ephemeris products provided by GFZ should be selected. In addition to the modeling day, the files of the previous day and the next day are also required.
Files_dcb	DCB products, in BSX format. (uncompressed)
Files_ionex	Reference ionospheric products provided by IAACs, in IONEX format.
	OUTPUT
OBS	Struct of *P* and *L* observations, in mat files.
SP3	Struct of satellite precise coordinates, in mat files.
P4	Struct of generated P4 observations, in mat files.
M_Result	Estimated ionospheric and DCB parameters, in mat files; Generated global and regional GIMs, in IONEX format.

M_Result_From_Test_Data	Results generated from Test_data set and Main_*.m programs
M_PLOT	Generated global and regional ionospheric figures, in fig or gif files.

3.4 Satellite selection

For different systems, we need to modify the "sate_mark.mat" file to select effective observable satellites and identify the on-orbit satellites. As shown in Fig. 1, the on-work satellites are marked as 1, and the invalid satellites are marked as 0. Users need to pay attention to the satellite update announcement by official institutions of each system.

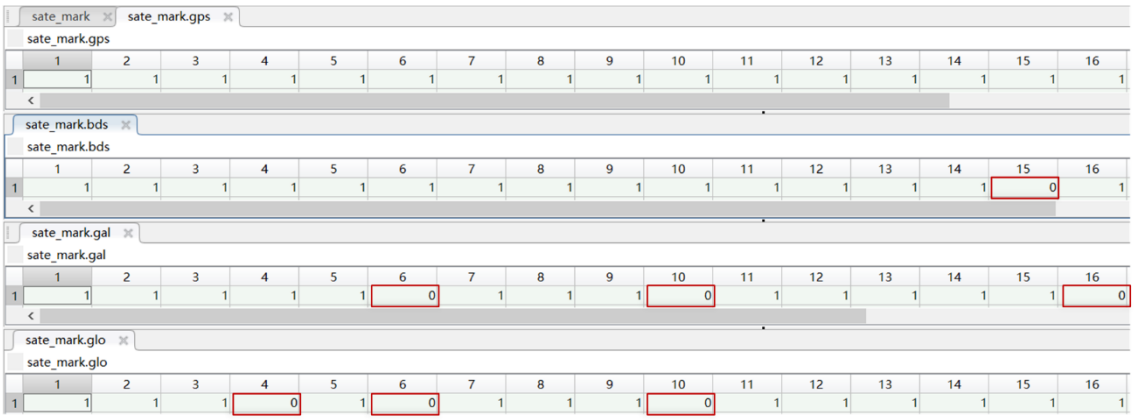


Fig. 1 Satellite selection of different systems

4. Processing strategy

M_GIM meanly includes global and regional ionospheric modeling, with spherical harmonics (SH) model for global ionospheric modeling, and polynomial model and nonintegral-SH model for regional ionospheric modeling. All of them have eight combinations to choose from, totally containing G, G+C, G+E, G+R, G+C+E, G+C+R, G+E+R and G+C+E+R, where G stands for GPS, C for BDS, E for Galileo and R for GLONASS. The main features of the M_GIM software are as follows:

- Support of global and regional ionospheric modeling, and multi-constellation selections based on GPS.
- Support of free setting of modeling cycle, cut-off angle and the weight of different system observations.
- Convenient visualization with M_Map (Pawlowicz, 2020) toolbox.
- Standard IONEX files (Schaer et al., 1998) generation.

Taking global ionospheric modeling as an example, the quad-system (G+C+E+R) combination is chosen. Along with the parameters estimated for VTEC modeling, the DCBs of GPS C1WC2W, BDS C2IC7I, GLONASS C1PC2P are estimated as by-products in the experiments, while for Galileo two types of DCBs C1CC5Q and C1XC5X are estimated respectively. The basic strategies and parameter settings for the data processing in the experiments are listed in Table 2.

Table 2 Basic strategies and parameter settings in the experiments

Items	Strategies

Systems	G+C+E+R
	GPS: C1W C2W L1C L2W
	BDS: C2I C7I L2I L7I
Frequencies	GALILEO: C1C C5Q L1C L5Q
	C1X C5X L1X L5X
	GLONASS: C1P C2P L1P L2P
Satellite orbits	Fixed to the GFZ MGEX products
Reference DCB value	Fixed to the CAS MGEX products
Cut-off angle	10°
The altitude of F_2 layer	450.00 km
Observation weight	G: 1 C: 1/2 E: 1 R: 1/3
DCB datum	Zero-mean condition
Ionospheric MFs	Single-layer MF
Estimator	Sequential LS method
Estimation	Satellite and receiver DCBs, ionospheric parameters

The three Main_*.m are the principal functions of global and regional ionospheric modeling. Three versions are provided, for the three different methods: spherical harmonics (SH) model for global ionospheric modeling, and polynomial model (POLY) and nonintegral-SH (nonSH) model for regional ionospheric modeling. Before executing any of the Main programs, the parameter configurations (Fig 2) can be modified. For quad-system global ionospheric modeling, **GCER** needs to be input in **system**. Note that the order of the GNSS system characters are hardcoded as options—for example, RECG is not valid, whereas GCER is valid for quad-system analysis. For other parameter configurations, **lim** refers to cut-off angle, **fig** refers to the number of ionospheric model parameter groups, **order** denotes the order of the spherical harmonic model, and **PG**, **PC**, **PE** and **PR** denote the weight of GPS, BDS, Galileo and GLONASS observations respectively. For users with Matlab's Parallel Computing Toolbox, **Corenum** denotes the cores number, which is commonly set as the maximum cores number of the current computer to improve computing efficiency.

```
%--system
% G : GPS-only   GC : G+C   GE : G+E   GR : G+R   GCE: G+C+E
% GCR: G+C+R   GER: G+E+R   GCER : G+C+E+R
system='GCER';
%--cut-off angle
lim=10;
%--number of ionosphere model parameter groups
fig=12;
%--SH model order and degree
order=15;
%--the weight of each system observation
PG=1; PC=1/2; PE=1; PR=1/3;
%--the cores number of computer processor
Corenum=2;
```

Fig. 2 Parameter configurations of global ionospheric modeling

The final estimation results are saved as a *.mat file under the **M_Result** folder. Model types will be saved separately, and are denoted by the file name of the result. For example, "GCER19275.mat" would be the SH solution for G+C+E+R analysis of data from year 2019, day of year 275.

Due to the low running efficiency of Matlab software and the huge amount of processing data, it will take a long time in ionospheric modeling. Runtimes can be improved by using the Parallel Computing Toolbox. The running time required under different processors and modeling schemes is listed in Table 3.

Table 3 Running time under different system and processor configuration

Main function	system	running time under different processor configuration (hour)	
		Core i5-6200U @2.30GHz (cores number: 2; threads number: 4)	Core i7-10875H @2.30GHz (cores number: 8; threads number: 16)
Main_SH.m	GCER	11.21	5.23
	G	5.78	2.30
Main_POLY.m	GCER	1.08	0.65
	G	0.55	0.31
Main_nonSH.m	GCER	1.19	0.66
	G	0.67	0.32

5. Plotting module

Before plotting the global and regional maps, a mapping package for Matlab called M_Map is needed, which can be downloaded from https://www.eoas.ubc.ca/~rich/map.html (Pawlowicz, 2020). The M_Map package must be found in the Matlab path; users can search the related online tutorials for help.

Note that the plotting scripts provided in the M_GIM distribution are written to display quad-system results. It is left to toolbox users to adapt the plotting scripts to display results with fewer systems.

5.1 Plot global and regional stations

Plot_SitesInf.m is the main function to plot the distribution of global and regional stations with quad-system results. Before running it, the estimated *.mat files under **M_Result** folder are needed after running "Main_*.m". The limits of the output map depend on the range set by **lat2**, **lat1**, **lon1** and **lon2**. The final pictures are saved in the **M_PLOT** folder in *.fig format.

```
%% plot station distributions with quad-system results
%--19: year; 275: day
figure;    doy=19275;
% Set latitude and longitude range
lat2=-90;      lat1=90;
lon1=-180;    lon2=180;
load(['M_Result/GCER',num2str(doy),'.mat']);
addpath('Tools/m_map','Tools/m_map/private');
[site_inf]=Plot_Multisites(lat2,lat1,lon1,lon2,G_R,C_R,E_R,EX_R,R_R,Sites_Info,list_P4);
saveas(gcf,['M_PLOT/','Global stations.fig'],'fig');
```

Fig. 3 Parameter configurations of the distribution of stations plotting

5.2 Plot ionospheric VTEC maps

Plot_*.m are the main functions to plot global and regional ionospheric maps with quad-system results. Plot_TEC.m is a basic function to plot the global and regional ionospheric VTEC maps. Take plotting global ionospheric VTEC maps as an example, the settings are shown in Fig. 4 and are also in Plot_SH.m.

```
%% ----------------- Plot and write global ionospheric VTEC maps --------------------
%% ================================================
doy=19275;  fig=12;
load('sate_mark.mat');
load(['M_Result/GCER',num2str(doy),'.mat']);
addpath('Tools/m_map','Tools/m_map/private');
lat2=-87.5;    lat1=87.5; lon1=-180;    lon2=180;
VVTEC = Get_VTEC(fig, 2.5, 5, IONC, NN, m0, 15);
VTEC=VVTEC;VTEC(VTEC(:,4)<0,4)=0.05;
RMS=[VTEC(:,1:3),VTEC(:,5)];
% % read IGS final GIMs (igsg2750.19i) as reference
IGSData=read_ionex(fig,'TEC');
DIFFTEC=[VTEC(:,1:3),VTEC(:,4)-IGSData(1:size(VTEC,1),4)];
Pname='2019-275-GCER- ';
Plot_TEC(fig,2.5,5,Pname,VTEC,lat1,lat2,lon1,lon2,0,50);
Pname='2019-275-GCER-D- ';
Plot_TEC(fig,2.5,5,Pname,DIFFTEC,lat1,lat2,lon1,lon2,-10,10);
Pname='2019-275-GCER-RMS- ';
Plot_TEC(fig,2.5,5,Pname,RMS,lat1,lat2,lon1,lon2,0,5);
% write results to ionex files
Write_ionex(doy,fig,G_R,C_R,E_R,EX_R,R_R,G_S,C_S,E_S,R_S,Pname,VTEC,sate_mark,total_station,list_P4);
```

Fig. 4 The settings in the global ionospheric VTEC maps plotting, see Plot_SH.m

As shown in Fig. 4, the **Get_VTEC** function aims to calculate the global ionospheric VTEC value; the **read_ionex** function aims to read the referenced IGS final GIMs; **plot_TEC** is the main function of plotting. Now users can run **Plot_SH.m** directly to get ionospheric VTEC maps, VTEC difference maps and RMS maps respectively, and the final plotting results are saved in the **M_PLOT** folder. The global ionospheric modeling results can be seen as follows:

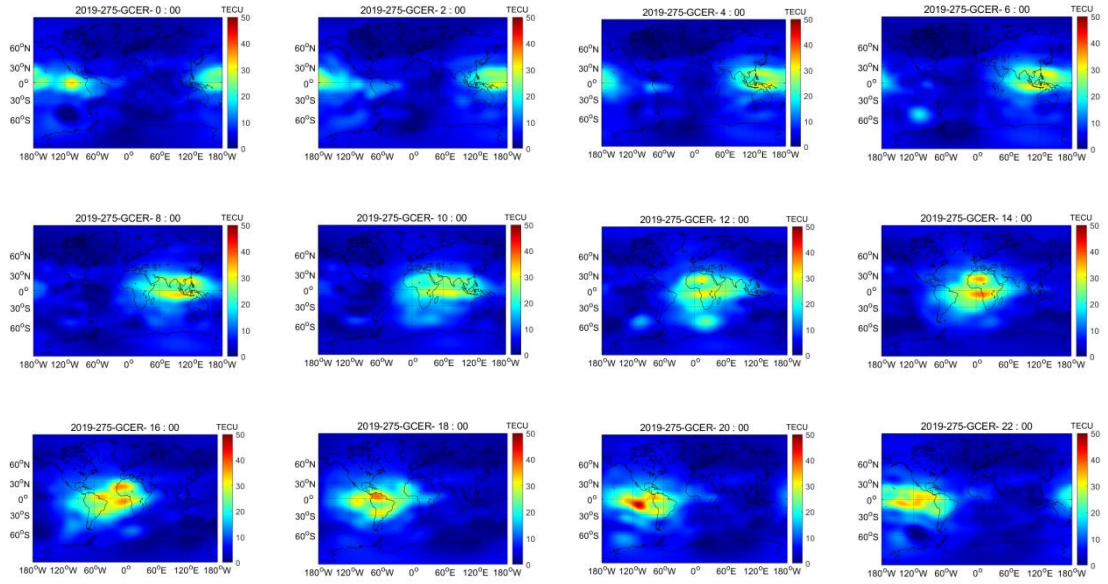


Fig. 5 The global VTEC maps

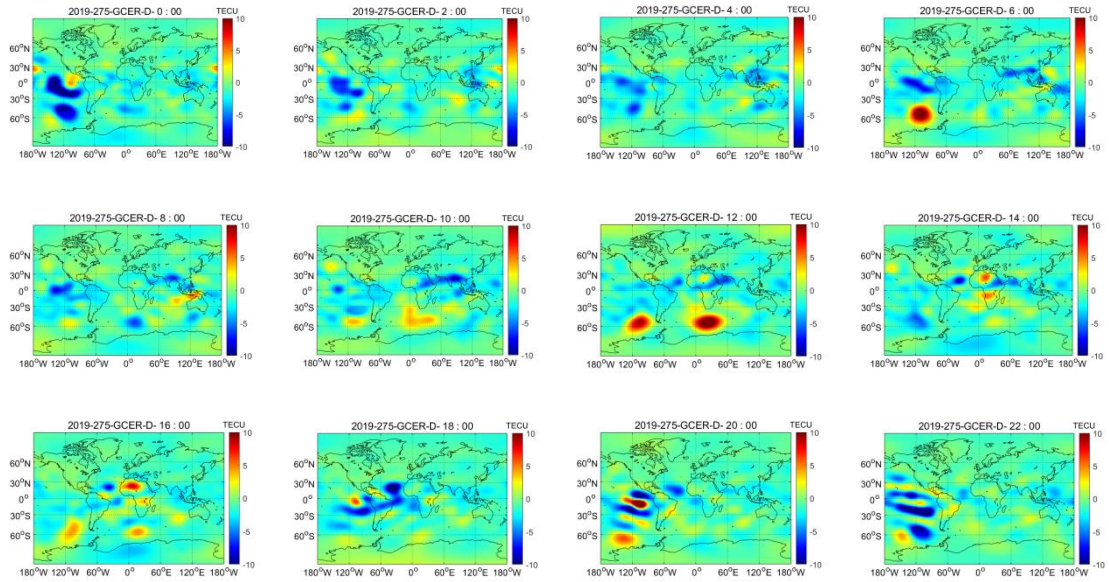


Fig. 6 The global VTEC difference maps compared with IGS final GIMs

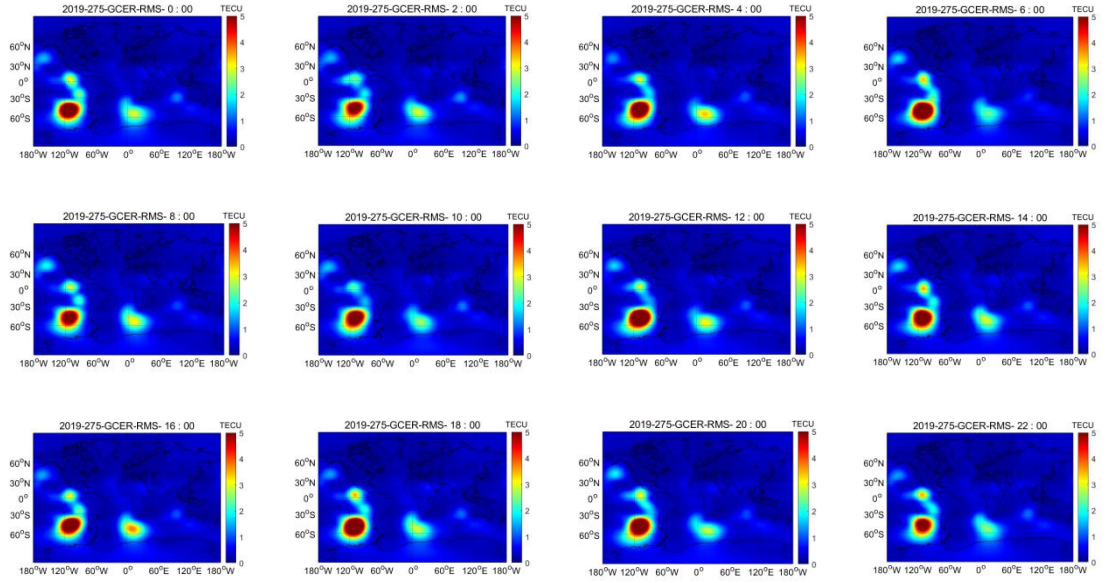


Fig. 7 The global VTEC RMS maps

6. Standard IONEX files generation

After plotting the ionospheric VTEC maps, the global or regional VTEC quad-system results can be written into a standard IONEX file by the **Write_ionex** function, and then be saved in the **M_Result** folder.

```
% write results to ionex files
Write_ionex(doy,fig,G_R,C_R,E_R,EX_R,R_R,G_S,C_S,E_S,R_S,Pname,VTEC,sate_mark,total_station,list_P4);
```

Fig. 8 The main function for writing IONEX files

Different standard IONEX files can be generated according to the area range and different time intervals. Some results of the generated global IONEX files are shown as follows:

```
     1                                                      START OF TEC MAP
  2019    10     2     0     0     0                        EPOCH OF CURRENT MAP
    87.5-180.0 180.0   5.0 450.0                           LAT/LON1/LON2/DLON/H
   27    27    28    28    28    28    28    29    29    29    29    28    28    28    28    28
   27    27    26    26    25    25    24    24    23    23    22    21    21    20    20    19
   19    18    18    18    17    17    17    17    17    16    16    16    16    16    17    17
   17    17    17    18    18    18    18    19    19    20    20    21    21    22    22    23
   23    24    24    25    25    26    26    26    27
    85.0-180.0 180.0   5.0 450.0                           LAT/LON1/LON2/DLON/H
   32    33    34    34    34    35    35    35    35    35    35    35    35    35    34    34
   33    32    31    30    29    28    27    26    25    24    22    21    20    19    18    17
   16    16    15    14    14    14    13    13    13    13    13    13    13    14    14    14
   14    15    15    15    16    16    17    17    18    18    19    20    21    22    23    24
   25    26    27    28    29    30    31    31    32
```

Fig. 9 Part of the generated IONEX files

7. Acknowledgement and support

This software is produced from M_DCB, and we are grateful to Jin et al. (2012) for their early work. We would like to thank the IGS, CODE, CAS, WHU, IGN, etc. for providing GNSS observations and products. Most of the data used in the experiments can be downloaded from the IGS data center of Wuhan University (ftp://igs.gnsswhu.cn/), the National Institute of Geographic and Forest Information Institute of France (ftp://igs.ign.fr/), as well as NASA's Crustal Dynamics Data Information System (ftp://cddis.nasa.gov/).

8. Disclaimer

Due to the time update and the continuous expansion of the satellites in the GNSS constellations, the "sate_mark.mat" needs to be reset accordingly. The experiment is carried out in Matlab 2019b environment. With the update of the software, proper modifications of the code may be needed. Some bugs may still exist in M_GIM, comments and suggestions from readers are welcome to send to the authors.

Appendix

MGEX observation files:

ftp://igs.gnsswhu.cn/pub/gps/data/daily/Year/DOY/**d

ftp://igs.ign.fr/pub/igs/data/campaign/mgex/daily/rinex3/Year/DOY

SP3 files:

ftp://igs.gnsswhu.cn/pub/gps/products/mgex/GPSWeek

ftp://igs.ign.fr/pub/igs/products/mgex/GPSWeek

DCB files:

ftp://igs.gnsswhu.cn/pub/gps/products/mgex/dcb/Year

ftp://igs.ign.fr/pub/igs/products/mgex/dcb/Year

IONEX files:

ftp://igs.gnsswhu.cn/pub/gps/products/ionex/Year/DOY

ftp://igs.ign.fr/pub/igs/products/ionosphere/Year/DOY

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